SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

CO5:Implementpolicy-basedreinforcementlearningmethodstodevelopefficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 35%

Dr G A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Mock Interview

1. Hierarchical Reinforcement Learning (HRL) – Autonomous Warehouse Navigation

- **AIM**

To simulate a Hierarchical Reinforcement Learning agent for autonomous warehouse navigation by dividing the navigation problem into smaller sub-tasks.

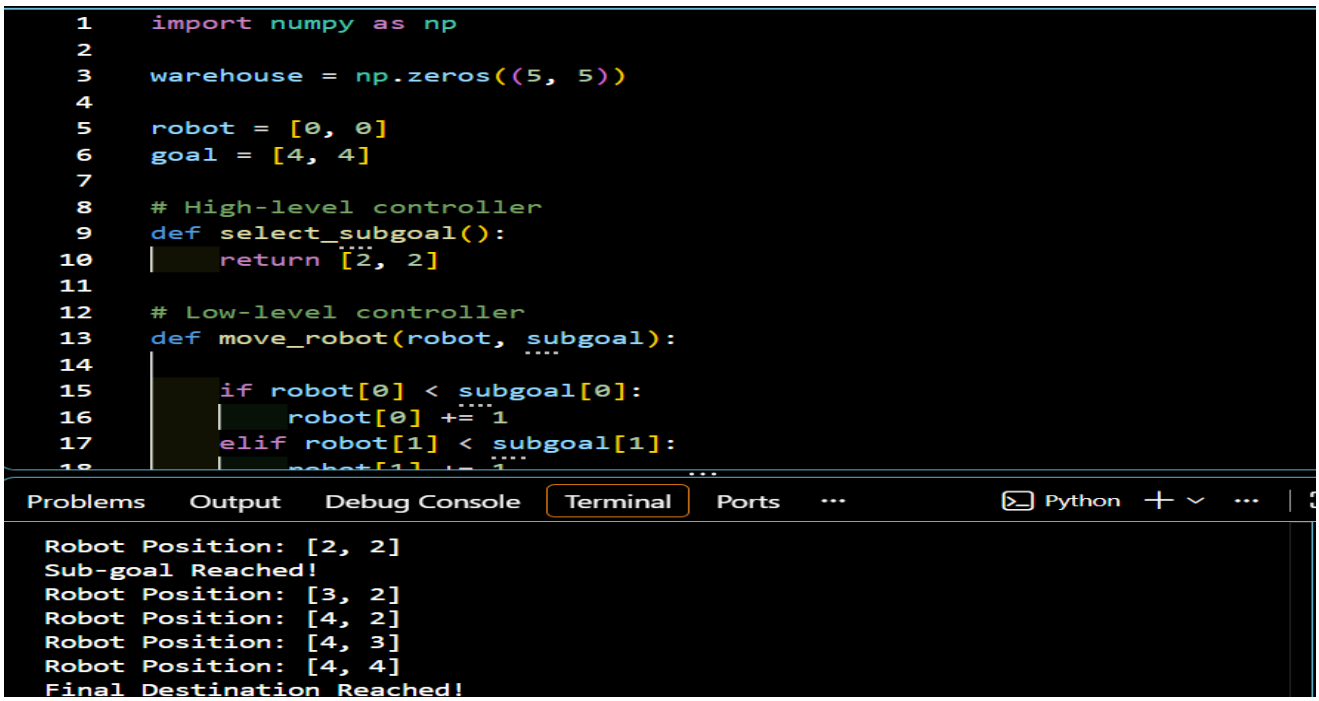
- **ABSTRACT**

Hierarchical Reinforcement Learning divides a complex task into high-level and low-level tasks. In a warehouse, the high-level agent decides the destination, such as selecting a storage location, while the low-level agent decides individual movements such as moving up, down, left, or right. This reduces the complexity of learning and helps the robot reach its destination efficiently.

- **ALGORITHM**

1. Create a warehouse grid environment.
2. Define the robot's initial position and target position.
3. Create a high-level controller to select the sub-goal.
4. Create a low-level controller to select movement actions.
5. Move the robot toward the selected sub-goal.
6. Provide a positive reward when the sub-goal is reached.
7. Provide a negative reward for obstacles or invalid movements.
8. Continue until the final destination is reached.
9. Display the learned navigation path.

- **OUTPUT:-**



CONCLUSION:-

The HRL agent successfully divides warehouse navigation into high-level and low-level tasks. This hierarchical approach simplifies complex navigation and helps the robot reach its destination efficiently.

2. HierarchicalAbstractMachine(HAM)–RobotSequentialTasks

AIM

ToimplementaHierarchicalAbstractMachineforcontrollingarobotthatperformssequentialtasks such as moving, picking an object, and delivering it.

ABSTRACT

A Hierarchical Abstract Machine represents robot behavior using a hierarchy of tasks and actions. A high-leveltaskisdividedintosmallersequentialoperations.Inthisproject,therobotfirstmovestoan

object, picks it up, moves to the delivery location, and finally drops the object.

ALGORITHM

1. Define the main robot task.
2. Divide the main task into smaller subtasks.
3. Move the robot to the pickup location.
4. Perform the pickup action.
5. Move to the delivery location.
6. Perform the drop action.
7. Check whether all tasks are completed.
8. Display the execution sequence.

OUTPUT:-

```
class RobotHAM:

    def move(self, location):
        print("Robot moving to:", location)

    def pick(self):
        print("Robot picked the object")

    def drop(self):
        print("Robot dropped the object")

    def execute(self):

        print("Main Task Started")

        # Sub-task 1
        self.move("Pickup Location")
```

```
Problems  Output  Debug Console  Terminal  Ports  ...  Python

anchigunashekar/OneDrive/Desktop/python-RL/CO5-2.PY
Main Task Started
Robot moving to: Pickup Location
Robot picked the object
Robot moving to: Delivery Location
Robot dropped the object
All Tasks Completed
```

CONCLUSION:-

TheHAMsuccessfullycontrolstherobotusinghierarchicalandsequentialtasks.Breaking the main task into smaller actions makes robot control easier to design, understand, and execute.

3. MAXQ Value Decomposition – Taxi Navigation

- **AIM**

To develop a simplified MAXQ Value Decomposition algorithm for a taxi navigation environment.

- **ABSTRACT**

MAXQisahierarchicalreinforcementlearningalgorithmthatdecomposesacomplextaskintosmaller subtasks. In a taxi environment, the main task of transporting a passenger can be divided into subtasks such as navigating to the passenger, picking up the passenger, navigating to the destination, and dropping off the passenger.

ALGORITHM

1. Define the taxi environment.
2. Define the main taxi task.
3. Divide the task into subtasks.
4. Calculate the value of each subtask.
5. Select the subtask with the highest value.
6. Execute the selected subtask.
7. Update the value of the subtask.
8. Repeat until the passenger reaches the destination.
9. Display the learned values.

OUTPUT:-

CONCLUSION:-

TheMAXQapproachdecomposestaxinavigationintosmallersubtasksandlearnsthevalueofeach task independently. This hierarchical decomposition makes complex navigation problems easier to solve.

4. Meta-Reinforcement Learning – Multiple GridWorld Environments

- **AIM**

TosimulateaMeta-ReinforcementLearningagentthatrapidlyadaptstodifferentGridWorld environments.

ABSTRACT

Meta-Reinforcement Learning enables an agent to learn how to learn across multiple tasks. In this project,theagentistrainedondifferentGridWorldenvironmentswithdifferentgoallocations.Instead of learning each environment completely from the beginning, the agent uses previous knowledge to

adapt quickly to a new environment.

ALGORITHM

1. Create multiple GridWorld environments.
2. Assign different goal positions to each environment.
3. Initialize shared knowledge for the agent.
4. Train the agent on multiple environments.
5. Store the learned information.
6. Introduce a new GridWorld environment.
7. Use previous knowledge to select actions.
8. Adapt the policy using a small number of steps.
9. Measure adaptation performance.
10. Compare learning speed across environments.

OUTPUT:-

CONCLUSION:-

TheMeta-RLagentusesknowledgeobtainedfrommultipleGridWorldtaskstoimproveadaptationto new environments. This allows the agent to learn new tasks faster than learning each task independently.

5. Multi-AgentReinforcementLearning(MARL)–CooperativeRobotPathPlanning

AIM

TodesignaMulti-AgentReinforcementLearningsysteminwhichmultiplerobotscooperatetofind efficient paths while avoiding collisions.

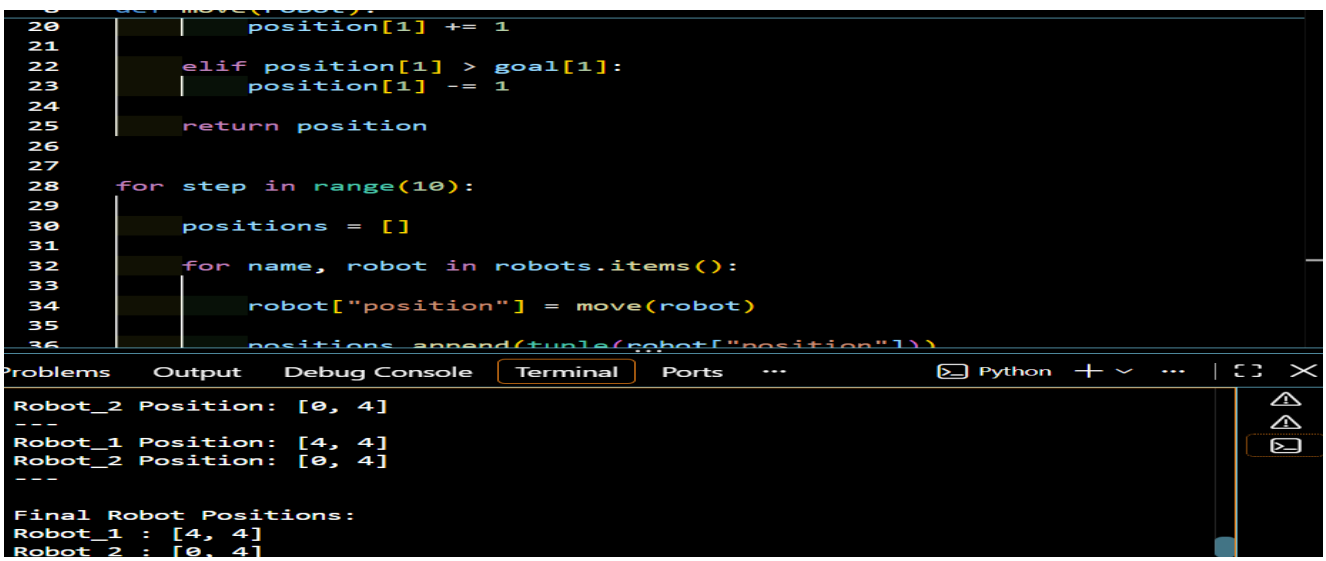
ABSTRACT

Multi-Agent Reinforcement Learning involves multiple agents learning and interacting within the sameenvironment.Incooperativerobotpathplanning,eachrobothasanindividualstartingposition anddestination.Therobotsshareinformationandrewardstoreachtheirdestinationswhileavoiding collisions and unnecessary movements.

ALGORITHM

1. Create a grid environment.
2. Initialize multiple robot agents.
3. Define the starting and goal positions.
4. Allow each robot to select an action.
5. Move the robots simultaneously.
6. Give positive rewards for reaching goals.
7. Give negative rewards for collisions.
8. Update each robot's policy.
9. Repeat the process for multiple episodes.
10. Evaluate the final paths of all robots.

OUTPUT:-



CONCLUSION

The MARL system allows multiple robots to learn and perform path planning cooperatively. By consideringotherrobots'positionsandusingrewardsandpenalties,thesystemcanreducecollisions and improve overall navigation efficiency.

Code of Conduct:

I E TEJAVARDHAN REDDY (192425297)certify that this submissionismyoriginal workandthatI have adhered to the guidelines specified for this assessment.
Iunderstand that any violation of academic integrity rules will result in disciplinary action.

Signatureofthestudent:

E TEJAVARDHAN REDDY

RUBRICSFOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accuratesubject knowledge with in-depth understanding	Showsgood knowledge with minor gaps	Basicknowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solvesproblems logically and appliesconcepts effectivelyinreal scenarios	Applies conceptswith moderate accuracy	Limitedproblem-solving ability; requires guidance	Unableto solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured,logical answers	Provides reasonable analysiswith some structure	Superficialor partially structured responses	Noanalytical thinking demonstrated
Communication	15	Communicates clearly, confidently,and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positiveattitude, andprofessional behavior throughout	Showsgood confidence with minor issues	Lowconfidence; inconsistent behavior	Lacks confidence and professionalism
Responseto Questions	10	Responds accurately, promptly, and elaborateswith relevant examples	Responds correctlybut with limited depth	Partial responses; needsprompting	Unable to respond appropriately